

MOVING CHARGES AND MAGNETISM

1. An electron (charge $-e$, mass ' m ') is revolving around a fixed proton in circular path of radius ' r '. The magnetic field at the centre due to electron is:

(1) 0
 (2) $\frac{\mu_0 e^2}{8\pi r^2 \sqrt{\pi m \epsilon_0} r}$
 (3) $\frac{\mu_0 e}{8\pi r \sqrt{\pi m \epsilon_0} r}$
 (4) $\frac{\mu_0 e}{4\pi r^2 \sqrt{\pi m \epsilon_0} r}$

2. A charged particle is projected in air horizontally near the earth's surface at a point where horizontal component of magnetic field is absent. Considering the effect of vertical component of the earth's magnetic field and the gravitational field due to the earth, the path of the particle will be :

- (1) circular (2) helical
 (3) straight line (4) parabolic

3. A long hollow tube is carrying an electric current along its length distributed uniformly on its circumference. The magnetic field:

- (1) increases linearly from the axis of the surface
 (2) is constant inside the tube
 (3) is zero only on the axis
 (4) is zero just out-side the tube.

4. For the magnetic field produced by a finitely long current carrying wire. Select wrong statement :

- (1) the lines of force will be concentric circles with centres on the wire.
 (2) There can be two points in the same plane where magnetic fields are same.
 (3) There can be large number of points where the magnetic field is same.
 (4) The magnetic field at a point is inversely proportional to the distance of the point from the wire.

5. Pick out the WRONG statement :

- (1) The gain in the K.E. of the electron moving at right angles to the magnetic field is zero.
 (2) When an electron is shot at right angles to the electric field, it traces a parabolic path.
 (3) An electron moving in the direction of the electric field gains K.E.
 (4) An electron at rest experiences no force in uniform and constant the magnetic field

6. The ratio of the energy density of magnetic field in the middle region of a solenoid to that in end region is :

- (1) 4 : 1 (2) 1 : 4 (3) 1 : 1 (4) 2 : 1

7. A charged particle is kept at rest on a smooth horizontal surface in uniform magnetic field ' B ' which is directed vertically downwards as shown in the figure.



An observer is moving with constant velocity v towards right. Then with respect to the observer which of the following statements is/are correct :

- (A) Path of the charged particle will be straight line.
 (B) Path of charged particle will be circular.
 (C) Magnetic force on the charged particle is zero.
 (D) Magnitude of magnetic force on charged particle is qVB .

Pick out the WRONG statement

- (1) A and D (2) A and C
 (3) B and D (4) B and C

8. Two parallel conductors carrying current in the same direction attract each other, while two parallel beams of electrons moving in the same direction repel each other. Which of the following statements cannot be the reason for this?

- (1) The conductors are electrically neutral
 (2) The conductors produce magnetic fields on each other
 (3) The electron beams do not produce magnetic fields on each other
 (4) The magnetic forces caused by the electron beams on each other are weaker than the electrostatic forces between them

9. In a cyclotron, a charged particle

- (1) undergoes acceleration all the time.
 (2) speeds up between the dees because of the magnetic field.
 (3) speeds up in a dee.
 (4) slows down within a dee and speeds up between dees.

10. The galvanometer consists of a coil, with many turns wound on a cylindrical soft iron core. When a current flows through the coil, a torque acts on it. This torque is given by $\tau = NIAB$. The magnetic torque $NIAB$ tends to rotate the coil by angle ϕ . A spring provides a counter torque $k\phi$ that balances the magnetic torque. The current sensitivity of the galvanometer is the deflection per unit current.
Statement-1 : Two galvanometers with identical construction but one with core of greater permeability will have lesser current sensitivity.
Statement-2 : Greater is the permeability of iron core greater is magnetic field.
- (1) Statement-I is true, Statement-II is true, Statement-II is the **correct** explanation of Statement-I
 (2) Statement-I is true, Statement-II is true, Statement-II is **not** the correct explanation of Statement-I.
 (3) Statement-I is **true**, Statement-II is false.
 (4) Statement-I is **false**, Statement-II is true.
11. To increase the current sensitivity of a moving coil galvanometer, we should :-
 (1) decrease number of turns in coil
 (2) decrease area of cross section of the coil
 (3) increase torsional constant of spiral springs
 (4) None of the above
12. Which of the following statement is correct ?
 (1) The magnetic field in the open space inside the toroid is constant
 (2) The magnetic field in the open space exterior to the toroid is constant
 (3) The magnitude of magnetic field inside the core of toroid is constant
 (4) The magnetic field inside the core of toroid is zero.
13. The nature of parallel and anti-parallel currents are :-
 (1) Parallel current repel and antiparallel currents attract
 (2) Parallel current attract and antiparallel currents repel
 (3) Both currents attracts
 (4) Both currents repel
14. The gyromagnetic ratio of an electron in sodium atom is :-
 (1) depending upon the atomic number of the atom
 (2) depending upon the shell number of the atom
 (3) independent of that orbit it is in
 (4) depending upon mass of atom
15. Magnetic field at the centre of a circular loop of area 'A' is 'B'. The magnetic moment of the loop is :-
 (1) $\frac{BA^2}{\mu_0\pi}$ (2) $\frac{BA\sqrt{A}}{\mu_0}$
 (3) $\frac{BA\sqrt{A}}{\mu_0\pi}$ (4) $\frac{2BA\sqrt{A}}{\mu_0\sqrt{\pi}}$
16. The order of magnetic field for human nerve fibre is :-
 (1) 10^{-2} (2) 10^{-10} (3) 10^{-12} (4) 10^{-5}
17. A field line is shown in fig. This field cannot represent :-

 (1) Magnetic field
 (2) Electrostatic field
 (3) Induced electric field
 (4) All of these
18. Tokamak is a device used for :-
 (1) accelerate the heavy charge particle
 (2) accelerate the light charge particle
 (3) plasma confinement in fusion power reactors
 (4) all
19. Gauss law of magnetism is :-
 (1) Net electric flux through any closed surface is zero
 (2) Net magnetic flux through any surface is zero
 (3) Net electric flux through any closed surface is (Q/ϵ_0)
 (4) Net magnetic flux through any closed surface is zero

20. If E and B denote electric and magnetic fields respectively, which of the following is dimensionless ?

(1) $\sqrt{\mu_0 \epsilon_0} \frac{E}{B}$

(2) $\mu_0 \epsilon_0 \frac{E}{B}$

(3) $\mu_0 \epsilon_0 \left(\frac{B}{E} \right)^2$

(4) $\frac{E}{\epsilon_0} \times \frac{\mu_0}{B}$

21. An irregular closed loop carrying a current has a shape such that the entire loop cannot lie in a single plane. If this is placed in a uniform magnetic field, the force acting on the loop :

(1) must be zero

(2) can never be zero

(3) may be zero

(4) will be zero only for one particular direction of the magnetic field